Question #1 of 60

Question ID: 1674564

Which of the following statements regarding the Black-Litterman model is *most accurate*?

A) Implied returns from reverse optimization are view adjusted and then used as inputs to mean variance optimization.

B) Implied returns from reverse optimization are used as inputs to mean variance optimization (MVO).

C) The Black-Litterman model starts with the global market portfolio's weights, which are assumed to be optimal.

Question #2 of 60

Question ID: 1580688

Daniel Roe and Loretta Morgan are both potential new clients of Grachek Investment Advisors. A summary of Ellen Grachek's notes concerning the potential clients are as follows:

- Roe stated that he wants to have a positive return on his $500,000 portfolio at all times, and that his required before-tax return is 5%. On a risk aversion questionnaire, Roe scored an 8, with 10 indicating the highest risk aversion.
- Morgan indicated that her $1,000,000 portfolio must generate a 2% return each year to meet her living expenses without making any withdrawals from the portfolio's principal. On a risk aversion questionnaire, Morgan scored a 3, with 10 indicating the highest risk aversion.

Grachek Investment Advisors has four model portfolios that they use for each client. Characteristics for each portfolio are identified as follows:

Portfolio	Expected Return	Standard Deviation
A	5.5%	7%
B	6.0%	8%
C	6.5%	10%
D	8.0%	15%

After reviewing her notes and making some calculations, Grachek makes the following statements regarding each client:

Statement 1:	Based on a utility-adjusted return of 2.54%, Portfolio B would be the best choice for Roe.
Statement 2:	Using Roy's safety-first measure, Portfolio D generates a score of 0.40, and would be the worst choice of the four for Morgan's portfolio.

Regarding Grachek's statements:

A) Statement 1 is incorrect; Statement 2 is correct.

B) Statement 1 is correct; Statement 2 is correct.

C) Statement 1 is incorrect; Statement 2 is incorrect.

Question #3 of 60

Question ID: 1587532

A portfolio manager expects that transaction costs will rise next year and that the correlation of equities with the rest of the portfolio will rise. What is the net impact of those changes on the portfolio's optimal rebalancing corridor width?

A) They will result in a wider corridor width.

B) They will result in a narrower corridor width.

C) They will result in an uncertain impact on corridor width.

Question #4 of 60

Question ID: 1580693

Melissa Brown, an analyst with Mollette Capital Advisors (Mollette), is reviewing the client profile of Karrie Jones. Mollette manages all of Jones's investment assets; however, because Brown is new to the firm, she has never met Jones. She does know, however, that Jones's asset allocation is appropriate given her age and investment policy statement. The allocation of Jones's portfolio is shown as follows:

Asset Class	Allocation (%)
Cash	5%
Intermediate-term Treasury bonds	0%
High-quality corporate bonds	5%
U.S. equities	60%
International equities	30%

Given Jones's asset allocation, which of the following conclusions about Jones is *most accurate*?

A) Jones has a large amount of financial capital.

B) Jones has a low risk tolerance.

C) Jones's human capital makes up the bulk of her portfolio.

Question #5 of 60

Question ID: 1551694

Which of the following asset classes is *least likely* to require a liquidity return premium?

A) Direct real estate.

B) Infrastructure.

C) Commodities.

Question #6 of 60

Question ID: 1580698

Which of these functions is *least likely* to be addressed with Monte Carlo simulation?

A) Assisting individual investors in identifying risk tolerance levels.

B) Rebalancing and taxes of a portfolio in a multiperiod framework.

C) Transforming a dataset with a nonnormal distribution to a normal distribution.

Question #7 of 60

Question ID: 1551722

A pension fund has pension assets that significantly exceed the value of pension liabilities. The fund wants to minimize the risk that the pension plan will become underfunded. Which of the following liability-relative approaches is *most appropriate* for the fund?

A) Integrated asset-liability approach.

B) Surplus optimization.

C) Two-portfolio (return-seeking) approach.

Question #8 of 60

Question ID: 1580685

The following information is available regarding corner portfolios from an efficient frontier.

Corner Portfolio	Expected Return	Exp. Std. Dev.	Sharpe Ratio	Asset Class Weights			
				1	2	3	4
1	9.90%	6.60%	1.197	0.00%	11.10%	88.90%	0.00%
2	10.06%	6.64%	1.214	0.00%	14.90%	85.10%	0.00%
3	12.10%	9.58%	1.054	55.22%	0.00%	44.78%	0.00%
4	14.23%	14.74%	0.830	100.00%	0.00%	0.00%	0.00%

An investor has a spending rate of 6%. If inflation is expected to be 4.50% annually and the cost of earning investment returns is 0.5%, what would be the utility of the portfolio that will, at a minimum, satisfy the investor's goals of capital preservation in real terms with a risk aversion value of 4?

A) 0.120.

B) 0.078.

C) 0.099.

Question #9 of 60

Question ID: 1587521

Which of the following statements regarding liabilities in the context of the asset allocation decision is correct?

A) A university endowment contribution is an example of a legal liability.

B) The pension liability associated with a defined pension plan is an example of a fixed liability.

C) A small pension liability in relation to the size of the sponsoring organization should be ignored.

Question #10 of 60

Question ID: 1551721

Which of the following statements regarding the two-portfolio (or the hedging/return-seeking) approach to liability-relative asset allocation is correct?

A) The asset allocation decisions regarding the two portfolios are made in conjunction with each other.

B) If the funding ratio is greater than 1, then it becomes difficult to create a hedging portfolio that completely hedges the liabilities.

C) The hedging portfolio could be constructed using cash flow matching.

Question #11 of 60

Question ID: 1587518

Which of the following methods is the *most appropriate* way of incorporating client risk preferences into asset allocations?

A) Specify additional constraints.

B) Specify a diversification objective.

C) Specify a risk tolerance factor.

Question #12 of 60

Question ID: 1587516

The following information is available regarding corner portfolios from an efficient frontier:

Corner Portfolio	Expected Return	Exp. Std. Dev.	Sharpe Ratio	Asset Class Weights				
				1	2	3	4	5
1	14.00%	18.00%	0.639	0.00%	0.00%	100.00%	0.00%	0.00%
2	13.66%	16.03%	0.696	0.00%	0.00%	86.36%	0%	14.00%
3	13.02%	13.58%	0.775	21.69%	0.00%	56.56%	0.00%	21.76%
4	12.79%	13.00%	0.792	21.48%	0.00%	52.01%	5.24%	21.27%
5	10.54%	8.14%	0.988	9.40%	51.30%	26.55%	0.00%	12.76%
6	8.70%	6.32%	0.981	0.00%	89.65%	4.67%	0.00%	5.68%

The following portfolios are under consideration by an investor:

Portfolio	Expected Return
A	11.0%
B	13.5%

For an investor with a risk-aversion of 6, which portfolio would have the highest utility?

A) Portfolio A with a utility of 0.092.

B) Portfolio A with a utility of 0.085.

C) Portfolio B with a utility of 0.115.

Question #13 of 60

Question ID: 1587523

Todd Zattau is the chief financial officer for the Crandall Steel Company, a mature U.S. steel processing company. The company provides a traditional defined benefit pension plan to all of its employees. The plan covers 5,000 employees, and the average age of workers who will eventually collect benefits is 52. Approximately 45% of the plan's participants are now retired and are receiving benefits. Zattau has hired Kara Rittenhouse, a financial advisor, to help him construct an IPS for the plan as well as recommend revisions to the plan's current investment allocation.

Zattau's progress on the IPS so far is shown here:

Return Objective	The discount rate applied to liabilities is 6.5%. Desired level of returns is 7.2%.
Risk Tolerance	?
Liquidity	?
Time Horizon	Company is a going concern, and new employees are still being added to the defined benefit plan, so the actual time horizon of the plan is infinite. However, the high percentage of retired participants and older workforce reduces the effective time horizon of the plan considerably.
Legal/Regulatory	Plan is subject to ERISA requirements.
Taxes	None
Unique Considerations	Plan is currently underfunded by 4%.

The current investment allocation for the plan is shown here:

Asset Class	Allocation (%)	Expected Return
Cash	3%	3.0%
Intermediate-term Treasury bonds	25%	5.0%
High-quality corporate bonds	32%	5.5%
U.S. equities	10%	8.5%
International equities	10%	10.0%
Venture capital	15%	19.0%

Based on the information provided, what is the risk tolerance for the Crandall Steel pension plan, and what should Rittenhouse recommend for the plan's allocation to cash and U.S.

equities, respectively?

	Risk Tolerance	Cash Allocation	U.S. Equities Allocation
A)	Below average	Lower	Lower
B)	Below average	Higher	Higher
C)	Above average	Adequate	Higher

Question #14 of 60

Question ID: 1622915

An institutional investor has a significant allocation to private equity in its portfolio. Which of the following approaches is *least appropriate* for addressing the liquidity risk of private equity in a mean-variance optimization (MVO) context?

A) Include private equity as an asset class when running MVO, but use the risk characteristics of that specific private equity investment instead of private equity as an asset class.

B) Include private equity as an asset class when running MVO using highly diversified asset class inputs.

C) Only exclude private equity as an asset class when running MVO.

Question #15 of 60

Question ID: 1551725

A bank is *most likely* to use which of the following approaches to liability-relative asset allocation?

A) Integrated asset-liability approach.

B) Two-portfolio approach.

C) Surplus efficient frontier approach.

Hikaru Tanaka is a portfolio manager with Yokohama Asset Management (YAM). Himari Nakamura, the Head of Risk at YAM, has been working with Tanaka to implement a revised risk budgeting approach that aims to maximize returns per unit of risk taken. Central to the

YAM risk budgeting approach is the adoption of two new risk metrics: the marginal contribution to risk (MCTR) and the absolute contribution to risk (ACTR).

Tanaka reviews an existing private client portfolio of Katashi Takahashi, who has recently contacted the firm with additional funds to invest. Tanaka decides to use the risk budgeting optimization approach to allocate these additional funds. He collects relevant data on the existing Takahashi portfolio, shown in Exhibit 1.

Exhibit 1: Takahashi Portfolio

Takahashi Portfolio Data	Beta	Expected Return	Weight
Domestic fixed income	0.359	3.25%	40%
Domestic equities	1.447	6.97%	32%
International equities	1.404	7.83%	28%
Domestic cash	0	2.50%	0%
Portfolio volatility		10.02%	

Question #16 - 19 of 60

Question ID: 1551708

Which asset class should be increased to improve the return/risk optimization of the Takahashi portfolio?

A) International equities.

B) Domestic equities.

C) Domestic fixed income.

Question #17 - 19 of 60

Question ID: 1551709

Which asset class should be reduced to move the Takahashi allocation closer to the optimal mix?

A) Domestic fixed income.

B) Domestic equities.

C) International equities.

Question #18 - 19 of 60

Question ID: 1551710

Which asset class of the Takahashi portfolio is *closest* to its optimal weighting?

A) Domestic equities.

B) Domestic fixed income.

C) International equities.

Question #19 - 19 of 60

Question ID: 1551711

Which of the following is an estimate of the percentage contribution of domestic bonds to overall portfolio risk of the Takahashi portfolio?

A) 13.36%.

B) 12.36%.

C) 14.36%.

Liam Brown is an entrepreneur of a start-up company that offers commission-based staffing services to U.S. corporate clients.

Brown, also a bodybuilding enthusiast, now plans to open a fitness club in 10 years. To do so, he needs to grow his $550,000 savings by 25%. Brown also needs to withdraw $20,000 annually to cover his children's education expenses. Brown meets his financial advisor, Jessica Miller, to allocate his assets. Miller uses mean-variance optimization (MVO) to choose an optimal asset mix for Brown. **Exhibit 1: Selected Information Pertaining to Three Asset Allocations and Brown's Objective Function** includes information pertaining to three asset allocations and Brown's objective function.

Exhibit 1: Selected Information Pertaining to Three Asset Allocations and Brown's Objective Function

Asset Allocation	LastYear's Return	Expected Return	Standard Deviation of Returns
Allocation A	5.10%	5.22%	9.20%
Allocation B	5.35%	6.02%	12.00%
Allocation C	5.95%	6.52%	13.00%

Brown's objective function is $U_m = E(R_m) - 0.02\sigma_m^2$, where U_m is Brown's utility for asset allocation *m*, and $E(R_m)$ is the expected return for asset allocation *m*. Note that $E(R_m)$ and σ_m are expressed as percentages rather than as decimals.

Miller meets Benjamin Jones, a long-time friend who is a portfolio manager at a large wealth management firm. When discussing Brown's case, Jones suggests the following:

Suggestion 1:	Force the optimizer to allocate 45% of Brown's tradable and nontradable assets to human capital.
Suggestion 2:	Allocate more funds to both non-U.S. equities and high-yield U.S. corporate bonds.

Brown's total net worth is shown in **Exhibit 2: Selected Information Pertaining to Brown's Total Net Worth (USD)**.

Exhibit 2: Selected Information Pertaining to Brown's Total Net Worth (USD)

Asset Class	Current Value
Financial assets (optimized)	550,000
Human capital	450,000
Inherited home	250,000
	1,250,000

Jones and Miller then discuss reverse optimization and resampling. Jones makes the following statements:

Statement 1:	Unlike reverse optimization, resampling addresses the quality issue of MVO inputs.
Statement 2:	To apply reverse optimization, the market-capitalization weights of the global market portfolio can be used as the starting optimal weights to imply the expected returns.

Jones reminds Miller to consider the risk-budgeting perspective when choosing an optimal asset allocation. He elaborates on this concept by providing Miller an illustrative example (**Exhibit 3: Illustrative Example About Risk Budgeting**). Jones then makes the following comments pertaining to the example:

Comment 1:	The strategic asset allocation is non-optimal from a risk-budgeting perspective.

Comment 2:	To make the strategic asset allocation more optimal, less funds need to be allocated to non-U.S. equities, and more funds need to be allocated to high-yield corporate bonds.

Exhibit 3: Illustrative Example About Risk Budgeting

Asset Class	Weight	Expected Return	Beta	Absolute Contribution to Total Risk	Percentage Contribution to Total Standard Deviation
Cash	10%	2.25%	0	0	0
U.S. equities	35%	5.64%	1.29	10.84%	45.17%
Non-U.S. equities	20%	6.01%	1.16	5.57%	23.21%
Investment-grade bonds	15%	2.65%	0.46	1.66%	6.92%
High-yield corporate bonds	20%	4.64%	1.24	5.95%	24.70%
	100.00%	**4.73%**		**24.00%**	**100.00%**

Question #20 - 23 of 60

Question ID: 1551703

Based upon the data in **Exhibit 1: Selected Information Pertaining to Three Asset Allocations and Brown's Objective Function** and the safety-first ratio, which of the following is *most likely* the optimal asset allocation that has the highest chance of meeting Brown's return goal?

A) Allocation C.

B) Allocation B.

C) Allocation A.

Question #21 - 23 of 60

Question ID: 1551704

Based on **Exhibit 2: Selected Information Pertaining to Brown's Total Net Worth (USD)** and Brown's occupation, which of the following is *most likely* correct?

A) Only Suggestion 2 is recommended.

B) Only Suggestion 1 is recommended.

C) Neither suggestion is recommended.

Question #22 - 23 of 60

Question ID: 1551705

Regarding the reverse optimization and resampling techniques, which of the following is *most accurate*?

A) Both statements are correct.

B) Only Statement 2 is correct.

C) Only Statement 1 is correct.

Question #23 - 23 of 60

Question ID: 1551706

Regarding the comments given by Jones, and using **Exhibit 3: Illustrative Example About Risk Budgeting**, which of the following is *most accurate*?

A) Both comments are correct.

B) Neither comment is correct.

C) Only Comment 1 is correct.

Question #24 of 60

Question ID: 1580701

A portfolio comprises three asset classes: 65% equities, 30% bonds, and 5% cash. The ratio of excess returns to the marginal contribution to total risk (MCTR) is equal to the portfolio's Sharpe ratio. This implies that:

A) The $MCTR_{equities}$ is equal to the absolute contribution of total risk, or $ACTR_{equities}$.

B) The 65%/30%/5% allocation is optimal from a risk-budgeting perspective.

C) The sum of the MCTRs of equities and bonds is equal to total portfolio risk.

Question #25 of 60

Question ID: 1580700

A portfolio comprises 70% Canadian equities and 30% Canadian bonds and has a standard deviation of 15%. The beta of Canadian equities is 1.2, while the beta of Canadian bonds is 0.65. What is the absolute contribution of total risk (ACTR) of Canadian equities?

A) 18.0%.

B) 12.6%.

C) 10.5%.

Question #26 of 60

Question ID: 1616953

The following information is available regarding corner portfolios from an efficient frontier.

Corner Portfolio	Expected Return	Exp. Std. Dev.	Asset Class Weights			
			1	2	3	4
1	6.90%	4.60%	0.00%	12.00%	88.00%	0.00%
2	10.00%	8.64%	0.00%	15.00%	45.00%	40.00%
3	13.00%	12.50%	55.00%	0.00%	45.00%	0.00%

A foundation has a spending rate of 8%. If inflation is expected to be 3.50% annually and the cost of earning investment returns is 0.005%, which of the following represents the correct weight of one of the asset classes that will, at a minimum, satisfy the investor's goals of capital preservation in real terms to an investor with a risk aversion value of 4?

A) Asset Class 4 with a weight of 19.6%.

B) Asset Class 2 with a weight of 7.7%.

C) Asset Class 1 with a weight of 27.7%.

Question #27 of 60

Question ID: 1580690

Shad Reed is on the board of trustees for the Wesley Ridge World Hunger Organization. The primary role of the organization is to oversee a large endowment fund that was originally established to provide food to low-income children in the United States. The original donor of the endowment recently died and provided the fund with another $200 million from his will to broaden the scope of the fund to provide food for hungry children all over the world. With the addition, the endowment's assets are currently valued at $600 million. When the fund was originally established, the spending rate was 5%; however, with the broader scope, the spending rate has increased to 6%. Also, because funds are going to be distributed to other countries, the board has determined that approximately 25% of the foundation's annual payout will be in the foreign currencies of other countries. The fund's revised investment policy statement is shown as follows:

Return Objective	Accounting for inflation of 2.5% and the new spending rate of 6%, the return requirement for the plan is 8.5%. A total return approach is appropriate.
Risk Tolerance	Above average, although risk tolerance has declined due to higher spending needs.
Liquidity	The endowment has minimal operating expenditures—liquidity requirements are low.
Time Horizon	Long-term
Legal/Regulatory	N/A
Taxes	N/A
Unique Considerations	N/A

The board has consulted with an investment advisor to discuss changes to the endowment's current asset allocation, which is shown as follows:

Asset Class	Allocation (%)	Expected Return	Expected Standard Deviation
Cash and marketable securities	7%	3.0%	2%
Intermediate-term U.S. Treasury bonds	28%	5.5%	7%
Foreign government bonds	8%	6.5%	10%
U.S. equities	45%	9.5%	18%
International equities	7%	11.0%	23%

Venture capital	5%	19.0%	38%

Which of the following recommendations would be *most appropriate* for the endowment fund?

A) Increase the allocation to foreign government bonds, increase the allocation to international equities, keep the allocation to cash the same, and keep the allocation to venture capital the same.

B) Decrease the allocation to U.S. equities, decrease the allocation to international equities, increase the allocation to foreign government bonds, and increase the allocation to intermediate-term U.S. Treasury bonds.

C) Increase the allocation to cash, decrease the allocation to U.S. equities, decrease the allocation to international equities, and increase the allocation to venture capital.

Question #28 of 60

Question ID: 1580687

The following information is a partial list of corner portfolios and asset class weights:

Portfolio	Exp. Return	Std. Dev.	Asset Class Weights		
			1	2	3
1	12.00%	10.50%	65.00%	0.00%	35.00%
2	16.50%	14.00%	15.00%	20.00%	50.00%
3	18.00%	20.00%	30.00%	20.00%	25.00%
4	23.00%	24.00%	15.00%	20.00%	55.00%

Which asset class is the most significant for an efficient portfolio with an expected return of 15% and the approximate standard deviation of this efficient portfolio?

A) Asset Class 1, 15.00%.

B) Asset Class 3, 12.85%.

C) Asset Class 2, 12.85%.

Question #29 of 60

Question ID: 1551733

Which of the following factors would *most likely* result in a narrower rebalancing corridor?

A) Higher transaction costs.

B) Higher volatility of asset classes.

C) Higher investor risk tolerance.

Question #30 of 60

Question ID: 1580699

Which of the following statements regarding risk and risk budgeting is correct?

A) Active risk is most relevant in an asset allocation implementation setting.

B) For risk budgeting purposes, risk can be defined as total risk, active risk, or misfit risk.

C) An optimal risk budget minimizes the total amount of portfolio risk that is allocated to the portfolio's constituent parts.

Question #31 of 60

Question ID: 1580692

Aaron Manning wishes to minimize the risk of his portfolio returns as measured by standard deviation while meeting a minimum expected return objective. The risk-free rate is 2%. The following asset allocations are available:

	Expected Return	Standard Deviation of Returns
Allocation 1	10%	12%
Allocation 2	7%	10%
Allocation 3	11%	15%

Based on the information provided, which of the following allocations should Manning choose?

A) Allocation 3.

B) Allocation 1.

C) Allocation 2.

Question #32 of 60

Question ID: 1587529

A 30-year-old wealthy investor wants to maximize her allocation to commodity investments. Which of the following approaches to asset allocation would *best* achieve her objective?

A) Endowment model.

B) 120 minus age rule.

C) 1/N rule, with 10 asset classes.

Question #33 of 60

Question ID: 1587527

Which of the following statements about an investor's goals-based asset allocation approach is *least correct*?

A) The investor's different goals with specified probabilities of success will be averaged, and a single module with the highest expected return will be used for the asset allocation.

B) The goals-based asset allocation approach typically uses pre-established subportfolios to meet client goals.

C) The asset opportunity sets will consist of both taxable and tax-exempt investments.

Question #34 of 60

Question ID: 1587520

Mean-variance optimization (MVO) outputs using factor exposures differ from using asset class exposures that reflect the same underlying assets in that:

A) their efficient frontiers differ from each other.

B) the choice of exposures reflects how an investor forms capital market expectations.

C) one approach will typically be superior to the other, depending on the space in which the investor operates.

Question #35 of 60

Question ID: 1580697

An institutional portfolio contains the following asset classes: direct real estate, infrastructure, and private equity. Which of the following statements is correct regarding the potential inclusion of those asset classes in performing mean-variance optimization (MVO)?

A) There are numerous indices available to accurately track such investments.

B) The risk-return characteristics of a specific investment in those assets are typically similar to the risk-return characteristics of the asset class.

C) Indices that track such investments are typically not investable.

Question #36 of 60

Question ID: 1587525

Taylor Robinson, age 60, recently retired from her position as director of public giving for United Electric Power, a large public utility company. Robinson has accumulated $2,000,000 in her 401(k) portfolio for retirement. Robinson estimates that she will need $50,000 after tax in today's dollars to live comfortably. Inflation is expected to be 2.5% annually. With her background in public giving, Robinson has two favorite charities and would like to make non-tax-deductible gifts of $10,000 to each of them annually, indexed for inflation. In her will, Robinson has specified that at her death, a gift fund will be established for each charity. Given this objective, one of Robinson's primary goals is to maintain the principal in her retirement fund to have a $1,000,000 gift account for each charity. Robinson recently met with her financial advisor, Brian Mitchell, CFA. During their meeting, Robinson stated, "If I wanted to gamble with my investments, I would play blackjack. At least then I would have fun losing money." Mitchell presented Robinson with three different model portfolios.

Asset Class	Portfolio A	Portfolio B	Portfolio C
Large-cap U.S. stocks	5%	15%	25%
Small-cap U.S. stocks	5%		15%
International—developed market equities	10%	5%	10%
International—emerging market equities	15%		5%
U.S. corporate bonds	10%	50%	30%
U.S. Treasury bonds	5%	20%	
Hedge fund of funds	10%		5%
Venture capital	15%		
Cash	25%	10%	10%
Total expected after-tax return	5.8%	5.5%	6.0%
Current yield	1.4%	1.6%	1.8%

Which of the portfolios would be *most appropriate* for Robinson?

A) Portfolio B.

B) Portfolio A.

C) Portfolio C.

Question #37 of 60

Question ID: 1587528

Which of the following heuristic and other approaches to asset allocation is *most closely* associated with the assumption of a lack of informationally efficient markets?

A) Norway model.

B) Endowment model.

C) 60/40 stock/bond heuristic model.

Question #38 of 60

Question ID: 1587531

In context of portfolio rebalancing, if the correlation of the asset class with the rest of the portfolio is lower, then the optimal corridor of the asset class will be:

A) narrower.

B) unchanged.

C) wider.

Question #39 of 60

Question ID: 1580682

Jeff Graefe has a risk aversion value of 6. He is evaluating three competing investments with the following characteristics. Which investment would have the *least* utility for Graefe?

Portfolio	Return	Std. Dev.
A	18.0%	24.0%
B	13.5%	10.0%
C	9.5%	6.0%

A) B.

B) A.

C) C.

Question #40 of 60

Question ID: 1580683

Dan Laske is evaluating three portfolios for investment of his retirement funds. Laske has a risk aversion value of 5. Which portfolio would be *best* for him?

Portfolio	Return	Std. Dev.
A	15.0%	17.0%
B	10.6%	10.0%
C	8.8%	8.0%

A) B.

B) A.

C) C.

Question #41 of 60

Question ID: 1587526

Which of the following statements regarding the goals-based approach to asset allocation is correct?

A) The relevant risk measure for both individuals and institutions is the probability of missing the goal.

B) Individuals are often concerned with single goals, while institutions are often concerned with multiple goals.

C) The return determination for individuals is based on minimum expectations, while for institutions, it is based on mathematical expectations.

Question #42 of 60

Question ID: 1587524

A risk manager at HIJ Bank is worried that an increase in interest rates will negatively impact the bank's surplus, while an increase in risk volatility may negatively impact some of its liabilities. Which liability-relative asset allocation approach would be *most appropriate* for the bank?

A) Two-portfolio approach.

B) Surplus optimization.

C) Integrated asset-liability approach.

Question #43 of 60

Question ID: 1580689

David Jalbert is considering three potential asset allocations. He wishes to earn a nominal return no less than 2% as he has a low risk tolerance with a lambda of 7.

The following asset allocations are available:

	Expected Return	Variance
Allocation 1	5%	0.0100
Allocation 2	7%	0.0143
Allocation 3	9%	0.0200

Based on the information provided, which of the following allocations should Jalbert choose?

A) Allocation 1.

B) Allocation 2.

C) Allocation 3.

Jonas Hoffman is an investment consultant for Wealth International Partners (WIP), located in Luxembourg. WIP specializes in providing investment and financial planning services for wealthy private European clients.

Hoffman has recently taken on three new clients and has asked his assistant, Klaus Salinger, to review their IPS objectives and constraints. These are summarized in Exhibit 1.

Exhibit 1: Client IPS Objectives and Constraints

New Clients	Elias Weber	Lina Wagner	Tobias Weisser
Investable assets	EUR 25.0 million	EUR 15.0 million	EUR 7.0 million
Required return	7.00%	5.50%	7.50%
Risk aversion coefficient	4.5	3.5	5.5
MVO constraints	—	—	No illiquid assets

Salinger also compiles a table of potential portfolio allocations, as shown in Exhibit 2. Salinger notes that Tobias Weisser is unable to invest in Allocations A, B, or C due to liquidity constraints.

Exhibit 2: Potential Portfolio Allocations

Allocations	Expected Return	Standard Deviation of Return
A	9.35%	16.00%
B	8.76%	14.78%
C	8.24%	14.25%
D	7.15%	12.64%
E	5.50%	10.32%
F	5.25%	10.11%
G	3.60%	7.28%
Risk free	1.75% per annum	

Question #44 - 47 of 60

Question ID: 1551688

For Elias Weber, for Portfolios B, D, and F, which of the following are ranked from highest to lowest in their utility (certainty equivalent return)?

A) F, B, D.

B) B, D, F.

C) D, F, B.

Question #45 - 47 of 60

Question ID: 1551689

For Lina Wagner, assuming Allocation A is recommended, and borrowing and lending at the risk-free rate is permitted, which of the following is the correct allocation to the risk-free rate?

A) EUR 7.60 million.

B) EUR 6.80 million.

C) EUR 7.40 million.

Question #46 - 47 of 60

Question ID: 1551690

For Tobias Weisser to achieve his required return of 7.5%, which of the following is *most correct*?

A) Lend EUR 0.412 million at the risk-free rate and invest EUR 7.412 million in Allocation D.

B) Borrow EUR 0.412 million at the risk-free rate and invest EUR 7.412 million in Allocation D.

C) Borrow EUR 0.454 million at the risk-free rate and invest EUR 7.454 million in Allocation D.

Question #47 - 47 of 60

Question ID: 1551691

Hoffman meets again with Lina Wagner, who advises that she now requires a minimum return of at least 2.5%, and a maximum volatility of 13%. Which of the following allocations should now be recommended to Wagner by Hoffman?

A) Allocation A.

B) Allocation E.

C) Allocation D.

Question #48 of 60

Question ID: 1580695

Consider an investor whose future wages are uncertain. The investor's human capital could be modeled as a(n):

A) mix of inflation-linked bonds, equities, and corporate bonds.

B) inflation-linked bond.

C) real return bond.

Question #49 of 60

Question ID: 1587530

The highest percentage allocation to fixed income for a 25-year-old investor would be under the:

A) 60/40 split.

B) 120 minus age rule.

C) 1/N rule, with 3 asset classes.

Question #50 of 60

Question ID: 1674562

A problem with mean variance optimization (MVO) is that:

A) it starts with portfolio weights from the global market portfolio, which are assumed to be optimal.

B) asset allocations are sometimes highly concentrated in a small number of asset classes while ignoring other asset classes.

C) an output is the implied returns, which must be view adjusted.

Question #51 of 60

Question ID: 1674563

Reverse optimization starts with:

A) expected returns, variances, correlations, a risk aversion factor, and constraints.

B) the global market portfolio's weights, which are assumed to be optimal.

C) view adjusted returns, which are then used as inputs to mean variance optimization (MVO).

Question #52 of 60

Question ID: 1580694

Andrew Tyson, age 31, has an existing investment portfolio consisting of investment-grade bonds and large-cap stocks. He also owns an apartment unit in the city and has a stable and promising career as an engineer. By including his apartment and his human capital in his investment portfolio, his capacity to bear risk would *most likely*:

A) decrease.

B) remain unchanged.

C) increase.

Question #53 of 60

Question ID: 1551674

Any mean-variance efficient portfolio has the:

A) lowest standard deviation and the highest expected return.

B) lowest standard deviation for a given level of expected return.

C) highest return among all other portfolios.

Question #54 of 60

Question ID: 1580691

Frances Bonner, a retired nurse, is a potential new client for Fullen Capital Management (Fullen). Bonner has told Fullen that to pay the living expenses not covered by her pension and Social Security, she must generate $10,000 annually on her $500,000 investment portfolio. She does not want to use principal to meet her living expenses. Fullen has four model portfolios that Bonner could use for her portfolio.

Portfolio	Expected Return	Standard Deviation
A	4.5%	6%
B	5.5%	8%
C	6.5%	10%

Based on Roy's safety-first measure, Bonner should select:

A) Portfolio A.

B) Portfolio B.

C) Portfolio C.

Question #55 of 60

Question ID: 1587517

Corri Morgan is an investment advisor for Izaguirre Investment Management (IIM). Morgan is reviewing the account for Brian and Nicole Herbster.

Brian and Nicole are both age 65, and have one daughter, Andrea, age 18. The Herbsters are recently retired from Tucker Technology, Inc., a large manufacturer of microprocessors for various applications. Andrea is an aspiring finance student and would like to attend a prestigious university to pursue a degree in finance. The tuition at the university costs $40,000 per year, but Andrea's strong academic performance in high school allowed her to earn a scholarship covering half of the tuition. The Herbsters have expressed a desire to fund the amount of the college education not covered by the scholarship, as well as leave a large inheritance to Andrea at their death. During their careers, the Herbsters earned relatively high incomes, and they were able to save approximately 10% of their income each year. Regarding their portfolio, they say they prefer investments that have minimal volatility. Their current investment portfolio is valued at $1.2 million.

The investment policy statement for the Herbsters is shown here:

Return Objective	The Herbsters' income requirement is $6,000 monthly. Total return requirement is $72,000 / $1,200,000 = 6%.
Risk Tolerance	Ability to take on risk: average. Willingness to take risk: below average.
Liquidity	Need cash each year for the next four years to fund college education.
Time Horizon	Three stages. Stage 1: funding daughter's college tuition. Stage 2: retirement. Stage 3, after death: inheritance for daughter.
Legal/Regulatory	N/A
Taxes	Little need to defer income.
Unique Considerations	Desire to fund daughter's college education.

The Herbsters' current portfolio is shown here:

Asset/Fund	Allocation (%)	Expected Return	Expected Yield	Expected Standard Deviation
Tucker Technology common stock	32%	19.0%	0.5%	28%
Money market fund	2%	2.5%	2.5%	2%
Diversified bond fund	30%	6.5%	5.5%	8%

Large capitalization equities	15%	9.5%	2.0%	16%
Emerging market equities	15%	16.0%	1.0%	26%
Undeveloped commercial land	6%	19.0%	0%	N/A

After reviewing the notes on the Herbsters, Morgan reviews recommendations complied by Todd Irons, a fellow portfolio manager with IIM. Irons's recommendations include the following:

Recommendation 1:	Reduce the weighting in Tucker Technology common stock—the large position exposes the portfolio to unnecessary security-specific risk.
Recommendation 2:	Increase the allocation to the diversified bond fund to increase income and decrease volatility.
Recommendation 3:	Increase the allocation to large capitalization equities to provide growth.
Recommendation 4:	Maintain the allocation to emerging market equities due to their high returns.
Recommendation 5:	Maintain the allocation to the undeveloped commercial land due to its low correlation with other assets in the portfolio.

After reviewing Irons's recommendations, Morgan should agree with:

A) Recommendations 1, 3, and 5 only.

B) Recommendations 1, 2, and 3 only.

C) Recommendations 1, 2, and 4 only.

Question #56 of 60

Question ID: 1551716

Which of the following statements about using investment factors in constructing asset allocations is *correct*?

A) The investment factors are market-neutral, long/short portfolios.

B) The use of investment factors is consistent with factor return models.

C) The factors are constructed to have high correlation with each other.

Question #57 of 60

Question ID: 1551724

The investment committee of a life insurance company recommends a strategic asset allocation for the company based on the projected policy premium inflows and payouts along with long-term capital market expectations. This approach to strategic asset allocation is known as the:

A) asset-liability approach.

B) static approach.

C) investment policy statement approach.

Question #58 of 60

Question ID: 1580684

Which of the following statements regarding mean variance optimization (MVO) is *least accurate*?

A) All asset weights add up to 100%.

B) Short positions are permitted.

C) An individual with average risk aversion will have a lambda of about 4.

Question #59 of 60

Question ID: 1587522

A pension plan wants to account for liability characteristics in its asset allocation decisions with respect to higher-than-expected payouts in the future. Which of the following liability characteristics *best* describes the plan's concerns?

A) Quasi-legal liabilities.

B) Legal liabilities.

C) Longevity risk.

Question #60 of 60

Question ID: 1587519

Which of the following statements regarding factor-based asset allocation is correct?

A) The factors may be highly correlated with the market portfolio.

B) The size factor is the combined return from a short position in small stocks and long position in large stocks.

C) The value-growth factor is the combined return from a short position in growth stocks and a long position in value stocks.